

UNCOVERING HUB GENES IN SHORT DURATION SPACEFLIGHT

Mervin June B. Gasatan¹, Wynn Dunn Gil D. Improso¹, and Ernest P. Macalalad¹

¹ Philippine Space Agency, Philippines
mervin.gasatan@philsa.gov.ph

Abstract. Spaceflight has diverse impacts on the human body. But due to the complexity of biological data, these changes are sometimes difficult to detect. Thus, in this study, network analysis is used to uncover important hub genes affected by short duration spaceflight to civilians. Differentially expressed genes from the transcriptome profiles of the Inspiration4 mission crew members were mapped in a network. And through degrees and centrality measures, hub genes were identified. Functional enrichment analysis (FEA) was done hereafter to understand their biological functions and were validated with other datasets in SOMA. Results uncovered eight hub genes namely UBA52, FAU, RACK1, RPS2, RPS11, RPS14, RPS15 and RPL35. FEA showed that these hub genes were all linked to ribosomal function. SOMA validation revealed that these ribosomal protein genes show consistent altered gene expression during spaceflight, suggesting potential links to interference in ribosome biogenesis and protein synthesis. These findings can help future studies in understanding the effect of spaceflight on humans.

Keywords: hub genes, spaceflight, network analysis

1 Introduction

Human spaceflight is one of the technological breakthroughs that mankind has achieved. Over the past 60 years since the first spaceflight, space missions are now increasing in duration, frequency, and extent, greatly exposing the human body to hostile conditions known to man [1]. It is known that spaceflight introduces physiological changes to the human body, including increased radiation exposure, bone density loss, skeletal muscle atrophy, and vision changes [2]. Unfortunately, the full extent of how the human body adapts to space environment is not fully understood up to the cellular level. There is a need to thoroughly understand human functioning and response in a more theoretically informed way [3]. One approach to this is analyzing the multi-omics research profiles.

Various research has already been done in understanding the different omics profile of astronauts, some of which are compiled in Space Omics and Medical Atlas (SOMA) [4]. Multi-omics data are usually massive and complex in nature, thus one way in uncovering important information is analyzing them as networks [5]. Amidst

the rich multi-omics data in SOMA, no recent studies were done in treating them as networks. This study is done to analyze the transcriptomic data of civilian astronauts in the Inspiration4 mission through network analysis. The crew members had no prior exposure to space but had undergone basic training for astronauts [6]. Findings can help uncover important genes affected by a short exposure of the human body to space. This study measures the degree and betweenness centrality of the protein-protein interaction (PPI) network to obtain the hub genes as hubs correspond to essential genes due to their involvement in more interactions making them more likely to be involved in one that is essential [7].

2 Methodology

2.1 Data Preparation of DEGs and PPI Network Construction

Transcription profiles of the astronauts' blood in the SpaceX Inspiration4 mission can be accessed in the NASA Open Science Data Repository in OSD-569 [6]. Differentially expressed genes (DEGs) dataset is accessible in the published study of Grigorev *et al.* [8] wherein the reads were aligned to the Gencode v41 human transcriptomic reference [9] with minimap2 version 2.24-r1122 [10] and processed further using different pipelines. Gene regulation changes at different timepoints before and after the spaceflight demonstrate the effect of spaceflight stress on the body at a molecular level.

DEG IDs were converted from Ensembl format to UniProt/SwissProt format through ID mapping in the UniProt server (<https://www.uniprot.org>) [11]. Non-coding genes were not included. Coding genes were then mapped in Search Tool for the Retrieval of Interacting Genes (STRING) v12.0 webserver (<https://string-db.org>) [12] for the construction of the protein-protein interaction (PPI) network. In the network, each node represents the protein and each edge as the protein interaction. The following filters were used to only select the relevant interaction: organism – *Homo sapiens*, required score – high confidence (0.700) and false discovery rate (FDR) stringency – 0.05. These filters ensure that the selected interactions are most likely true and relevant [11]. As for the FDR adjusted p-value, this limits the probability of false positives to be selected. The PPI is then exported to Cytoscape v3.10.3 for network analysis.

2.2 Hub Genes Identification and Functional Enrichment Analysis

Hub genes are known to be highly connected to the other genes in the network thus highlighting their importance [13]. The cytoHubba v0.1 plug-in [14] was used to identify the hub genes based on the degree and betweenness measure. Degree centrality measure is simply the measure of the number of connected nodes to the node of interest. While betweenness centrality shows the number of times a node is present in the shortest path with the neighboring nodes. The top 30 nodes of each measure were

analyzed, and the identified hub genes were selected if they were present in the top rankings of the two metrics.

Functional enrichment analyses (FEA) of the Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) terms were done on the identified hub genes through the Shiny GO v0.82 web server [15]. Results can link these gene lists to underlying molecular pathways and functional categories such as biological processes (BP), cellular component (CC) and molecular function (MF). Terms are only considered significant when FDR value is less than 0.05.

2.3 Validation of Hub Genes

Expression of hub genes were validated in publicly available international astronaut biobank, the Space Omics and Medical Atlas (SOMA) [4]. As the I4 mission was one of the major contributors in SOMA, the other datasets like the NASA Twin Study [16] and JAXA cell free RNA database [17] were referenced.

3 Results and Discussion

3.1 DEGs Identification and PPI Construction

The DEGs dataset were mined from the study of Grigorev *et al.* [8] on the direct RNA sequencing of civilian astronauts' blood. As such, 437 DEGs were identified. ID mapping in the UniProt server revealed 332 coding and 105 non-coding genes. Only the coding genes were used to construct the PPI network in STRING webserver. The STRING database was constructed from the integrated protein-protein interactions that include both physical interaction and functional association [12]. The exported PPI network from STRING to Cytoscape, as shown in Figure 1, showed a network with 207 nodes and 1568 edges. Disconnected nodes and interactions with scores less than 0.7 were not included in the final network.

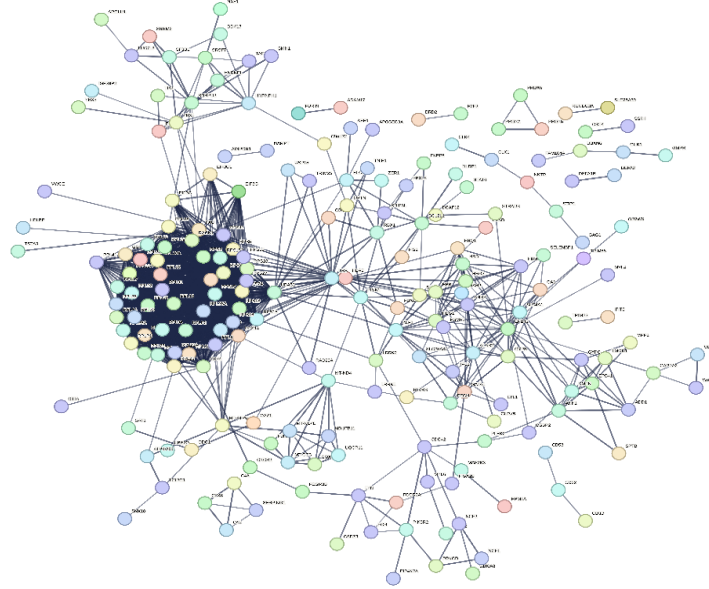


Fig. 1. PPI network of the DEGs expressed in the astronaut's blood in the I4 mission.

3.2 Hub Genes Identification and Functional Enrichment Analysis

To identify important proteins in the interactome, the two centrality measures (degree and betweenness) were used to find the hub genes. Analysis from the cytoHubba plug-in shows that there are 8 identified hub genes (UBA52, FAU, RACK1, RPS2, RPS11, RPS15, RPS14, RPL35) that are present in both the top 30 ranked genes based on degree and betweenness score as shown in Table 1.

Table 1. Identified Hub Genes Present in the Top 30 Degree and Betweenness Measure

Name	Degree Rank	Degree Score	Betweenness Rank	Betweenness Score
UBA52	1	57	1	5733.13
FAU	2	55	10	2224.53
RACK1	2	55	12	2108.15
RPS2	4	54	19	1162.47
RPS11	4	54	20	1156.49
RPS15	7	53	7	2826.23
RPS14	11	52	23	1013.13
RPL35	21	49	26	920.39

Functional enrichment analysis is a widely used method to discover biological annotations that are overrepresented in a list of genes with respect to a reference, usually the whole database [18]. Results for the FEA of hub genes related to short duration spaceflight are summarized in Figure 2. Figure 2A showed the significant GO BP, GO CC, and GO MF terms. Enriched GO BP terms are translation, peptide metabolic process, peptide biosynthetic process, cytoplasmic translation and amide biosynthetic process. As for the enriched GO CC terms, they are ribosome, ribosomal subunit, cytosolic ribosome, small ribosomal subunit and cytosolic small ribosomal subunit. While the enriched GO MF terms are structural molecule activity, structural constituent of ribosome, RNA binding, nucleic acid binding and mRNA binding. As for the significant KEGG pathways (Figure 2B), they mainly include ribosome and coronavirus diseases – COVID-19. As the main mechanism of COVID-19 is hijacking the host ribosome for its own viral protein translation thus utilizing some of the same proteins in the ribosome pathway [19], this significant KEGG pathway term will not be discussed further in this paper. The enriched GO and KEGG terms showed that all identified hub genes are related to ribosomal proteins (RPs), components of either the 40S or 60S ribosomal subunits. The importance of the role of ribosomal proteins on early spaceflight stress is still not fully understood thus needs to be studied further.

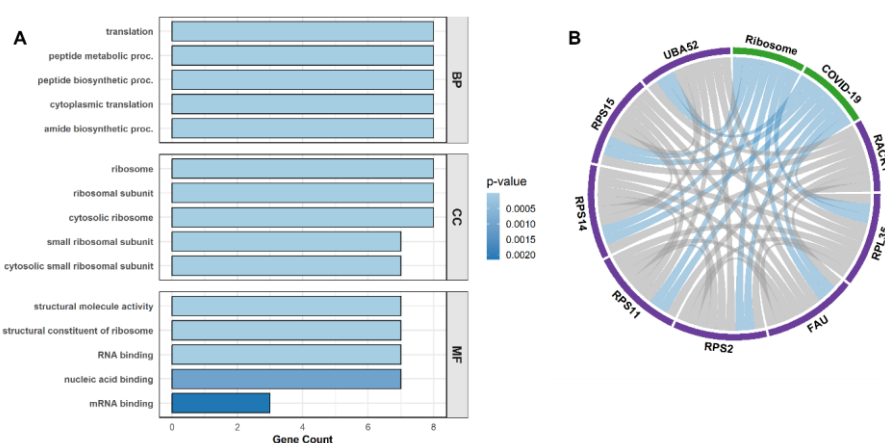


Fig. 2. Functional Enrichment Analysis of Hub Genes. (a) Bar plot of the enriched GO terms in relation to BP, CC and MF. (b) Chord diagram of significant KEGG pathways.

3.3 Validation of Hub Genes and their Significance to Spaceflight

The expression of hub genes was cross-referenced across the three spaceflight datasets (I4 Data, NASA Twins Study, JAXA CFE Study) found in SOMA and it is interesting to note that the hub genes were consistently differentially expressed across all studies. This is quite significant as each mission spans different periods of time (I4 – 3 days, JAXA – 120 days, NASA Twins – 340 days). Knowing that RPs show consistent transcriptional alterations in the space environment, understanding its impact

on human health is important especially with the increasing trend of duration of space missions across the years [20].

The impact of RPs altered expression on ribosome biogenesis has yet to be studied fully, more specifically in the context of spaceflight. Disruption of ribosome biogenesis can occur due to either increased or decreased expression of different ribosome components, and they are linked to promoting cell cycle arrest, senescence, or apoptosis [21]. This interference is usually associated with cancer, aging, and age-related degenerative diseases. Recent studies have already shown that the space environment is an accelerator of biological ageing [22] [23]. With the continuous altered expression of RPs in spaceflight, this may also be an overlooked link to accelerated ageing which needs further studying. On another note, interference in synthesis of ribosomes is also associated with cancer [24] [25]. Further research needs to be done to understand the role of RPs in cancer progression, especially in the context of spaceflight. It is already known that spaceflight increases the risk of cancer [1]. Many studies have already shown the effect of spaceflight on ribosome biogenesis [26] [27], but its significance may still not be highlighted enough. Furthermore, it must be noted that under stress situations, decreased ribosome activity resulting in reduced protein synthesis, and subsequent growth arrest were already observed [24]. Thus, transcriptional alterations of RPs suggest links of spaceflight in the interference of ribosome biogenesis and protein synthesis.

The study successfully uncovered hub genes (UBA52, FAU, RACK1, RPS2, RPS11, RPS14, RPS15, RPL35) related to the short-term exposure of civilians in space. Functional enrichment analysis shows that these hub genes are related to the ribosome. Cross-validation with the other datasets in SOMA reveals that the ribosomal proteins are consistently differentially expressed during spaceflight even for short periods of time. These findings suggest that spaceflight alters the expression of ribosomal proteins and the implications of this connection to human health need to be studied further.

References

1. Muhsen, I. N., Zubair, A. C., Niederwieser, T., Hashmi, S. K. Space exploration and cancer: The risks of deeper space adventures. *Leukemia* 38(9), 1872–1875 (2024).
2. Ong, J., & Lee, A. G. An introduction to space medicine and the physiological effects of spaceflight on the human body. In *Spaceflight Associated Neuro-Ocular Syndrome*, pp. 1–7. Academic Press (2022).
3. Sarma, M. S., Shelhamer, M. The human biology of spaceflight. *American Journal of Human Biology* 36(3) (2024).
4. Overbey, E.G., Kim, J., Tierney, B.T. *et al.* The Space Omics and Medical Atlas (SOMA) and international astronaut biobank. *Nature* 632, 1145–1154 (2024).
5. Tolani, P., Gupta, S., Yadav, K., *et al.* Big Data, integrative OMICS and Network Biology. *Advances in Protein Chemistry and Structural Biology* 127–160 (2021).
6. OSD-569, <https://osdr.nasa.gov/bio/repo/data/studies/OSD-569>, last accessed 2025/05/05.

7. Goymer, P. Why do we need hubs? *Nature Reviews Genetics* 9(9), 651–651 (2008).
8. Grigorev, K., Nelson, T.M., Overbey, E.G. et al. Direct RNA sequencing of astronaut blood reveals spaceflight-associated m6A increases and hematopoietic transcriptional responses. *Nat Commun* 15, 4950 (2024).
9. Frankish, A. et al. GENCODE 2021. *Nucleic Acids Res.* 49, D916–D923 (2021).
10. Li, H. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics* 34, 3094–3100 (2018).
11. Ahmad S, da Costa Gonzales L J, Bowler-Barnett E H, *et al.* UniProt Consortium The UniProt website API: facilitating programmatic access to protein knowledge. *Nucleic Acids Research*, gkaf394 (2025).
12. Szklarczyk D, Kirsch R, Koutrouli M, et al. The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Res.* 51(D1): D638–646 (2023).
13. Li, S., Nakayama, H., & Sinha, N. R. How to utilize comparative transcriptomics to dissect morphological diversity in plants. *Current Opinion in Plant Biology* 76, 102474 (2023).
14. Chin, C.-H., Chen, S.-H., Wu, H.-H., *et al.* Cytohubba: Identifying hub objects and sub-networks from complex interactome. *BMC Systems Biology* 8(S4) (2014).
15. Ge, S. X., Jung, D., Yao, R. ShinyGO: A graphical gene-set enrichment tool for animals and plants. *Bioinformatics* 36(8), 2628–2629 (2019).
16. Garrett-Bakelman, F. *et al.* The NASA Twins Study: A multidimensional analysis of a year-long human spaceflight. *Science* 364, eaau8650 (2019).
17. OSD-530, <https://osdr.nasa.gov/bio/repo/data/studies/OSD-530>, last accessed 2025/05/14
18. Garcia-Moreno, A., López-Domínguez, R., Villatoro-García, J. A., *et al.* Functional Enrichment Analysis of regulatory elements. *Biomedicines* 10(3), 590 (2022).
19. Zhang, D., Zhu, L., Wang, Y., *et al.* Translational control of covid-19 and its therapeutic implication. *Frontiers in Immunology* 13 (2022).
20. Smith, M. G., Kelley, M., Basner, M. A brief history of spaceflight from 1961 to 2020: An analysis of missions and astronaut demographics. *Acta Astronautica* 175, 290–299. (2020).
21. Turi, Z., Lacey, M., Mistrik, M., Moudry, P. Impaired ribosome biogenesis: Mechanisms and relevance to cancer and aging. *Aging* 11(8), 2512–2540 (2019).
22. Campisi, M., Cannella, L., Pavanello, S. Cosmic Chronometers: Is spaceflight a catalyst for biological ageing? *Ageing Research Reviews* 95, 102227 (2024).
23. Kuznetsov, N. V., Statsenko, Y., Ljubisavljevic, M. An update on neuroaging on Earth and in spaceflight. *International Journal of Molecular Sciences* 26(4), 1738 (2025).
24. Wang, W., Nag, S., Zhang, X., *et al.* Ribosomal Proteins and Human Diseases: Pathogenesis, Molecular Mechanisms, and Therapeutic Implications. *Medicinal Research Reviews* 35(2), 225–285 (2014).
25. Ruggero D, Pandolfi PP. Does the ribosome translate cancer? *Nat Rev Cancer* 3(3):179–192 (2003).
26. Kamal, K. Y., Herranz, R., van Loon, J. J., Medina, F. J. Simulated microgravity, Mars gravity, and 2G hypergravity affect cell cycle regulation, ribosome biogenesis, and epigenetics in *Arabidopsis* cell cultures. *Scientific Reports* 8(1) (2018).
27. Basu, P., Kruse, C. P. S., Luesse, D. R., Wyatt, S. E. Growth in spaceflight hardware results in alterations to the transcriptome and Proteome. *Life Sciences in Space Research* 15, 88–96 (2017).